Bayes' Theorem

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Flood Data

Data of monthly rainfall in Kerala from 1901 to 2018 is provided.

Attributes in the Dataset:

```
YEAR: Year between 1901 to 2018

JAN, FEB,..., DEC:Rainfall in mm in the specified month

ANNUAL RAINFALL: Total rainfall in mm in that year

FLOODS: YES or NO, in that year
```

Our objective is to utilize Bayes' Theorem to compute the conditional probability of July rainfall less than 500 millimeters, given the absence of flooding in the same year?

Assuming that:

- A: Rainfall in July less than 500 mm
- B: Flood

P(A|not B) = P(not B|A) x P(A)/P(not B)

Import necessary libraries, load the 'kerala_floods.csv' dataset, display a data sample, determine dataset dimensions, and identify any missing values.

In [5]:
```
#Import all required libraries
import pandas as pd
import numpy as np
```

In [6]:
```
#Read and show file 'kerala_floods.csv'
df = pd.read_csv('kerala_floods.csv')
df.head()
```

Out[6]:

	SUBDIVISION	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL RAINFALL	FLOODS
0	KERALA	1901	28.7	44.7	51.6	160.0	174.7	824.6	743.0	357.5	197.7	266.9	350.8	48.4	3248.6	YES
1	KERALA	1902	6.7	2.6	57.3	83.9	134.5	390.9	1205.0	315.8	491.6	358.4	158.3	121.5	3326.6	YES
2	KERALA	1903	3.2	18.6	3.1	83.6	249.7	558.6	1022.5	420.2	341.8	354.1	157.0	59.0	3271.2	YES
3	KERALA	1904	23.7	3.0	32.2	71.5	235.7	1098.2	725.5	351.8	222.7	328.1	33.9	3.3	3129.7	YES
4	KERALA	1905	1.2	22.3	9.4	105.9	263.3	850.2	520.5	293.6	217.2	383.5	74.4	0.2	2741.6	NO

In [7]:
```
#Show the size
df.shape
```

Out[7]:
```
(118, 16)
```

In [8]:
```
#Check if you have missing values
df.isnull().sum()
```

Out[8]:
```
SUBDIVISION          0
YEAR                 0
JAN                  0
FEB                  0
MAR                  0
APR                  0
MAY                  0
JUN                  0
JUL                  0
AUG                  0
SEP                  0
OCT                  0
NOV                  0
DEC                  0
 ANNUAL RAINFALL     0
FLOODS               0
dtype: int64
```

To start, we'll prepare the columns needed for our analysis. This will help us work with the columns more efficiently.

Create a new column entitled 'July_less_than_500' and assign a value of 0 of rows with July rainfall exceeding 500 millimeters, and 1 to those with rainfall less than or equal to 500 millimeters.

In [11]:
```
# create new column with the name "July_less_than_500" and assign 1 for more 500 and 0 for less 500
df['july_less500'] = (df['JUL']<500).astype('int') #if less than 500, it will give one, other than that 0
```

Assign 1 for yes and 0 for no in 'FLOODS' column

In [13]:
```
df['FLOODS'] = df['FLOODS'].map({"YES":1,"NO":0})
df.head()
```

Out[13]:

	SUBDIVISION	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL RAINFALL	FLOODS	ju
0	KERALA	1901	28.7	44.7	51.6	160.0	174.7	824.6	743.0	357.5	197.7	266.9	350.8	48.4	3248.6	1	
1	KERALA	1902	6.7	2.6	57.3	83.9	134.5	390.9	1205.0	315.8	491.6	358.4	158.3	121.5	3326.6	1	
2	KERALA	1903	3.2	18.6	3.1	83.6	249.7	558.6	1022.5	420.2	341.8	354.1	157.0	59.0	3271.2	1	
3	KERALA	1904	23.7	3.0	32.2	71.5	235.7	1098.2	725.5	351.8	222.7	328.1	33.9	3.3	3129.7	1	
4	KERALA	1905	1.2	22.3	9.4	105.9	263.3	850.2	520.5	293.6	217.2	383.5	74.4	0.2	2741.6	0	

As the analysis will be confined to the 'YEAR', 'july_less500', and 'FLOODS' columns, we will get the dataset based on these columns.

In [15]:

```
#To answer our question we may select only the columns that will help to do so
df2 = df[['YEAR','july_less500','FLOODS']]
df2.head()
```

Out[15]:

	YEAR	july_less500	FLOODS
0	1901	0	1
1	1902	0	1
2	1903	0	1
3	1904	0	1
4	1905	0	0

Sugggestion: Cross tabulation / Frequancy distrbution table will be effective to calculate the probabilites that you need her.

In [17]:

```
#make a cross tabulation for 'FLOODS' and 'July_less_than_500'
pd.crosstab(df2['FLOODS'],df2['july_less500'],margins=True)
```

Out[17]:

july_less500	0	1	All
FLOODS			
0	39	19	58
1	57	3	60
All	96	22	118

P(A)

In [19]:
```
# Find the probability of rainfall less than 500mm in July
p_A = 22/118
p_A
```

Out[19]: 0.1864406779661017

P(not B)

In [21]:
```
#Find the probability of not Flood
p_not_B = 58/118
p_not_B
```

Out[21]: 0.4915254237288136

P(not B and A)

In [23]:
```
#Find the prob of not flood and rainfall less than 500mm in July
p_not_B_AND_A = 19/118
p_not_B_AND_A
```

Out[23]: 0.16101694915254236

P(not B | A) = P (not B and A) / P(A)

In [25]:
```
# Find the probability of not_flood given it rained less than 500mm
p_not_B_given_A = p_not_B_AND_A/p_A
```

```
p_not_B_given_A
```

Out[25]: 0.8636363636363635

P(not A)

In [27]:
```
# Find the probability of rainfall not less than 500mm in July
p_not_A = 96/118
p_not_A
```

Out[27]: 0.8135593220338984

P(not B|not A) = P(not B and not A)/P(not A)

In [29]:
```
# Find the probability of not flood given it rained not less than 500mm
p_not_B_given_not_A = 39/96
p_not_B_given_not_A
```

Out[29]: 0.40625

What is probability of having a rainfall less than 500mm in month of July, provided it has not flooded in that year ?

P(A|not B) = P(not B|A) x P(A)/P(not B)

Answer 1: Use Bayes' throrem only

In [32]:
```
#Option 1: Using Bayes' throrem only
p_A_given_not_B = p_not_B_given_A*p_A/p_not_B
p_A_given_not_B

#p_not_B = (p_A*p_not_B_given_A)+ (p_not_A*p_not_B_given_not_A)
```

Out[32]: 0.32758620689655166

Answer 2: Use Bayes' throrem with the law of total probability

In [34]:
```
#Option 2: Using Bayes' throrem with the law of total probability
p_A_given_not_B = (p_not_B_given_A*p_A)/((p_A*p_not_B_given_A)+ (p_not_A*p_not_B_given_not_A))
p_A_given_not_B
```

Out[34]:
```
0.32758620689655166
```

In [ ]: